TITLE PAGE

Problem Statement ID: **SIH26263**

Problem Statement Title: Development of an APAAR ID-Based Student Data Management and Verification Platform for AICTE-Approved Institutions

Theme: Smart Automation

PS Category: Software

Team ID: [YOUR TEAM ID]

Team Name: **TCE_Yquanto (Thiagarajar College of Engineering)**

Live Repository: **https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM**

Prototype Domain: https://pramaan-aicte.yourdomain.com [PASTE CUSTOM DOMAIN HERE]

TCE_Yquanto

PRAMAAN-AICTE: APAAR-BASED REGULATORY RECONCILIATION & MONITORING

SMART INDIA HACKATHON 2026

Proposed Solution & Architectural Overview

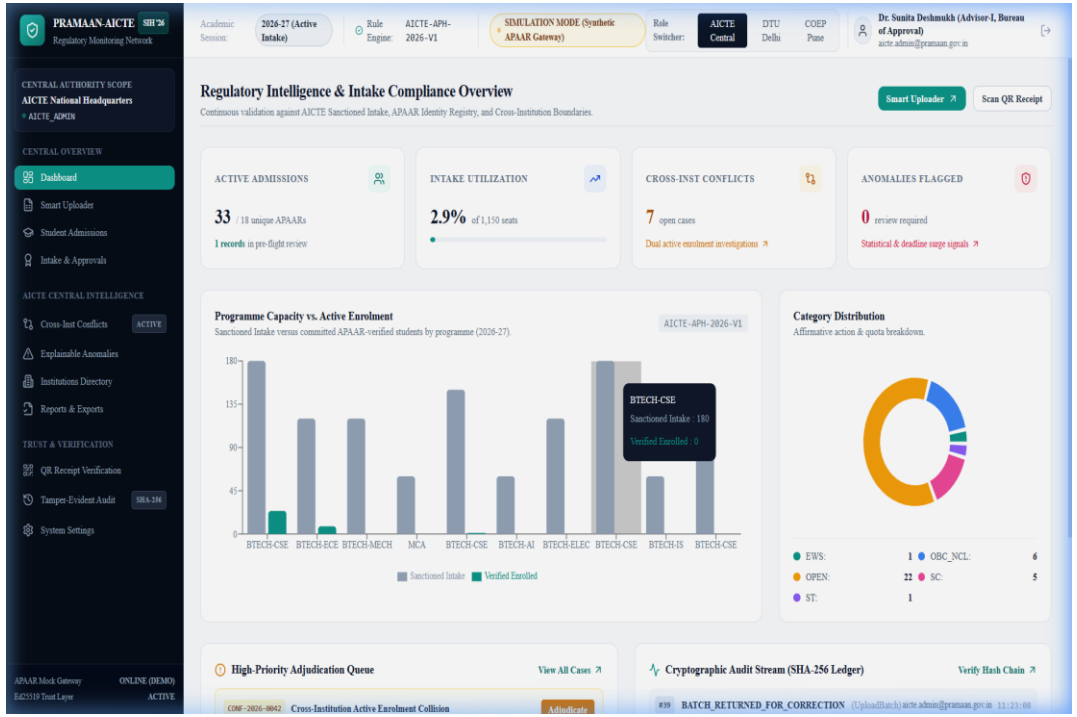
- **Dual-Portal Ecosystem:** Decouples institutional ingestion workspaces from the AICTE Central Executive Command Center for tenant privacy.
- **Automated Batch Lifecycle:** Excel/CSV upload -> Smart Header Alias Normalization -> Pre-flight Sandbox -> Atomic 2-Phase Commit.
- **Real-Time Regulatory Audit:** Continuous validation ensures complete institutional record authenticity before official academic census locking.

Addressing Core Problem Vulnerabilities

- **Ghost Admissions Elimination:** Anchors all technical admissions to verified 12-digit national APAAR IDs validated via Verhoeff dihedral D5 permutations.
- **Intake Ceiling Enforcement:** Validates admissions against versioned AICTE Approval Handbook (APH 2026-27) limits (General, TFW 5%, EWS 10%, NRI/FN 15%).
- **Cross-College Collision Radar:** Real-time cross-institutional duplicate detection prevents fraudulent concurrent multi-college technical admissions.

Key Technical Innovations & Security Layer

- **Zero Plaintext Storage:** Keyed HMAC-SHA256 indexing + dynamic masking (XXXX-XXXX-9012) satisfies DPDP Act 2023 without searchable plaintext exposure.
- **Asymmetric Verifiability:** Issues Ed25519 (Curve25519) digitally signed verification receipts with high-density QR codes (zero PII in public URLs).
- **Tamper-Evident Ledger:** Sequential SHA-256 cryptographic hash chain guarantees non-repudiation and detects single-bit database modifications.



PROTOTYPE EVIDENCE & LIVE VERIFICATION

AICTE National Command Center (/dashboard) featuring live sanctioned vs verified intake metrics, affirmative action quotas, duplicate conflict alerts, and real-time audit ledger feeds.

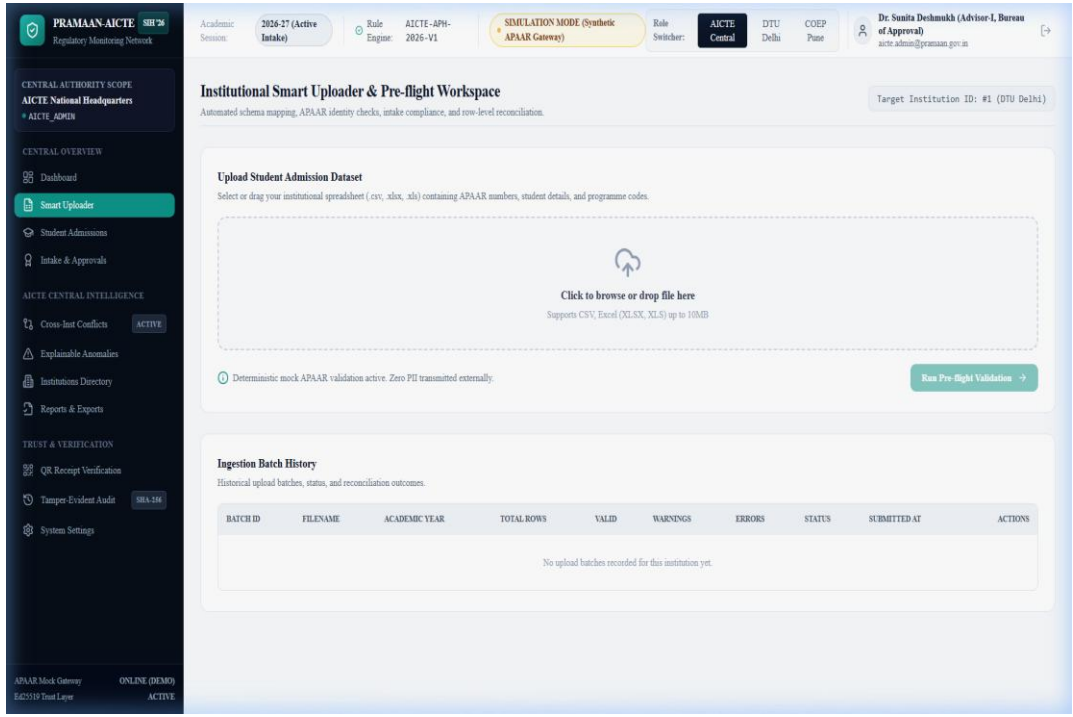
TECHNICAL APPROACH & DECOUPLED GOVERNANCE ARCHITECTURE

Core Production Technology Stack

- **Frontend Architecture:** Next.js 14 App Router, TypeScript (Strict), Tailwind CSS, Lucide Icons, Recharts Analytics.
- **Backend Micro-Services:** Python 3.10+, FastAPI (Modular Monolith), Pydantic v2, Uvicorn, OpenAPI 3.1 REST contracts.
- **Persistence & Cloud DB:** Supabase Cloud PostgreSQL (Port 6543 pooler, RLS policies) / SQLite WAL zero-setup local dev.
- **Security & Cryptography:** Ed25519 asymmetric signatures (PyNaCl/RFC 8032), Keyed HMAC-SHA256, bcrypt password hashing.
- **Resilient Ingestion Core:** Pandas & OpenPyXL with regex normalizer for fault-tolerant tabular parsing across diverse ERP schemas.

5-Tier Implementation Methodology & Pipeline

- **1. Resilient Ingestion Desk:** Sub-50ms fuzzy alias matching resolves inconsistent college column headers (e.g., student_apaar <-> apaar_id).
- **2. Pre-flight Validation Gate:** Quarantines rows; executes Verhoeff identity syntax, approved intake bounds, and supernumerary quota isolation.
- **3. Atomic Staging Sandbox:** Enables in-browser spreadsheet inspection, error highlighting, and inline corrections prior to permanent submission.
- **4. Regulatory Adjudication:** Automated double-enrollment radar triggers dispute hearings with recorded reviewer remarks and college notices.
- **5. Cryptographic Proof Engine:** Issues Ed25519 signed verification receipts and appends tamper-evident SHA-256 audit ledger events.



PROTOTYPE EVIDENCE & LIVE VERIFICATION

Smart Ingestion Desk (/uploads) showcasing automated schema normalizer, pre-flight error severity categorization (ERROR, WARNING, INFO), inline cell editing, and atomic batch commitment.

FEASIBILITY, SCALABILITY & RISK MITIGATION FRAMEWORK

Feasibility Analysis & Scalability Metrics

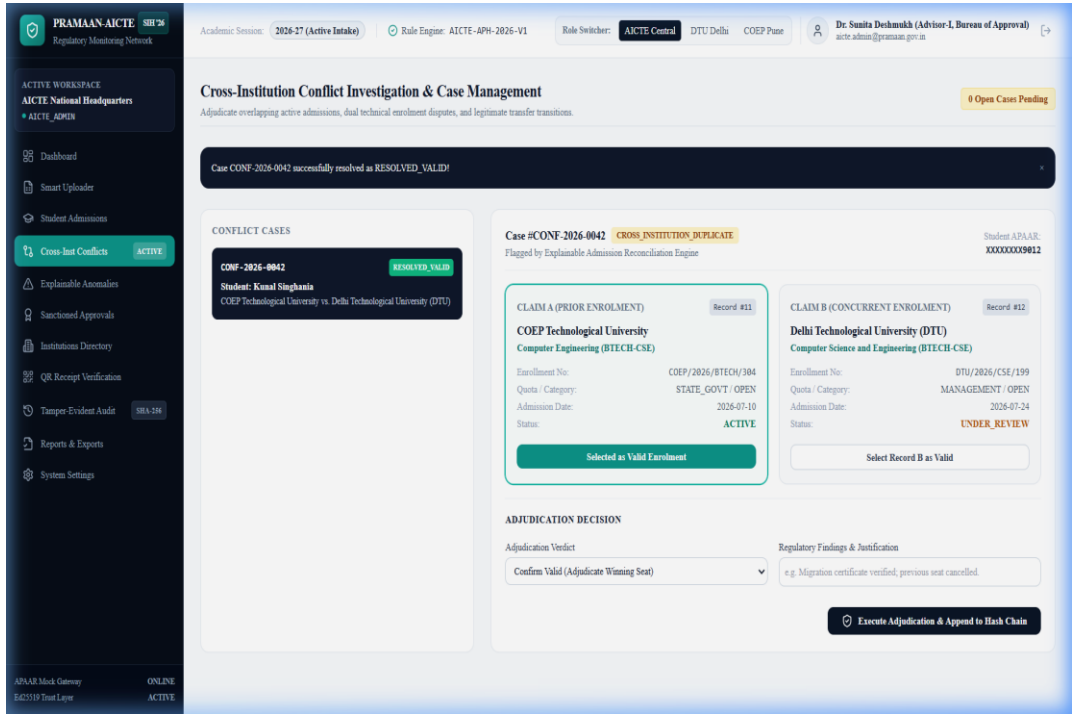
- **Operational Feasibility:** Zero client installation; operates seamlessly over standard institutional web browsers on any modern OS.
- **Verified Prototype:** Exhaustively verified with 59/59 passing pytest test suite and 33/33 live verified FastAPI endpoints.
- **High-Throughput Ingestion:** Chunked asynchronous database commits process 5,000+ student batches in under 2 seconds.

Identified Governance Challenges & Risks

- **Heterogeneous College Formats:** Thousands of approved colleges maintain conflicting spreadsheet schemas, date layouts, and branch codes.
- **Identity Privacy Risks:** Plaintext storage or logging of 12-digit student APAAR identifiers violates DPDP Act 2023 mandates.
- **Cross-Institution Collisions:** Simultaneous student admission claims across multiple colleges create jurisdictional deadlocks.
- **Database Tamper Risks:** Risk of illicit database modification of sanctioned intake quotas or student admission statuses.

Concrete Technical Countermeasures

- **Fuzzy Normalization Engine:** Regex alias dictionary automatically standardizes column headers and auto-detects date formats.
- **Privacy-Preserving Hashing:** HMAC-SHA256 indexing + dynamic masking guarantees zero plaintext APAAR exposure across the cloud.
- **Side-by-Side Adjudication:** Automated dispute queue with side-by-side claim timestamps, fee receipts, and one-click resolution.
- **Cryptographic Hash Chain:** Sequential SHA-256 hash chaining mathematically exposes any historical record tampering.



PROTOTYPE EVIDENCE & LIVE VERIFICATION

Cross-Institution Duplicate Adjudication Console (/conflicts) demonstrating Case CONF-2026-0042 (DTU Delhi vs COEP Pune) with side-by-side claim timestamps, fee proofs, and resolution controls.

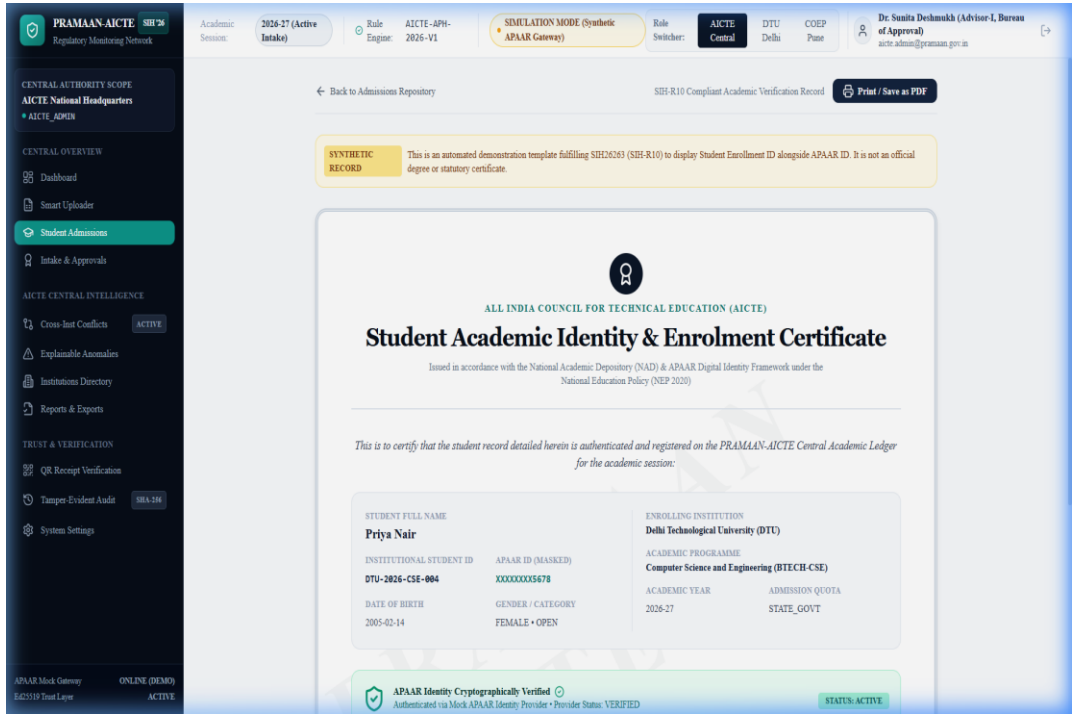
REGULATORY IMPACT, SOCIAL EQUITY & SIH-R10 COMPLIANCE

Impact on Target Regulatory Stakeholders

- **AICTE & Regulators:** Complete nationwide visibility over technical admissions; eliminates ghost seats and manual verification backlogs.
- **Institutions & Colleges:** Eliminates physical documentation overhead; reduces annual student admission auditing from 3 months to 5 minutes.
- **Students & Parents:** Guarantees authentic seat reservation; transparent supernumerary quota allocation protects deserving students.
- **Employers & Verifiers:** Sub-second zero-trust academic verification via mobile QR scan without contacting institutional registrars.

Quantifiable Multi-Dimensional Benefits

- **Economic Benefit:** Stops ghost-seat grant diversion and illegal seat-blocking, saving crores in fraudulent scholarship reimbursements.
- **Social Equity:** Strictly safeguards affirmative action quotas (TFW 5%, EWS 10%, KM, NRI/FN 15%) from unauthorized reallocation.
- **NEP 2020 & DPDP Alignment:** Full alignment with National Academic Depository (NAD), DigiLocker, and DPDP Act 2023 privacy mandates.
- **Environmental & Operational:** 100% paperless verification workflow eliminates courier delays and physical document falsification.
- **SIH Requirement 10 Compliance:** Built-in Degree Certificate Generator displaying Student ID alongside Masked APAAR ID and Ed25519 QR.



PROTOTYPE EVIDENCE & LIVE VERIFICATION

SIH-R10 Compliant Degree Certificate Engine (/admissions/certificate/[id]) displaying Student Roll No, Masked APAAR ID, Council Seal, and Asymmetric Ed25519 QR Code for instant offline verification.

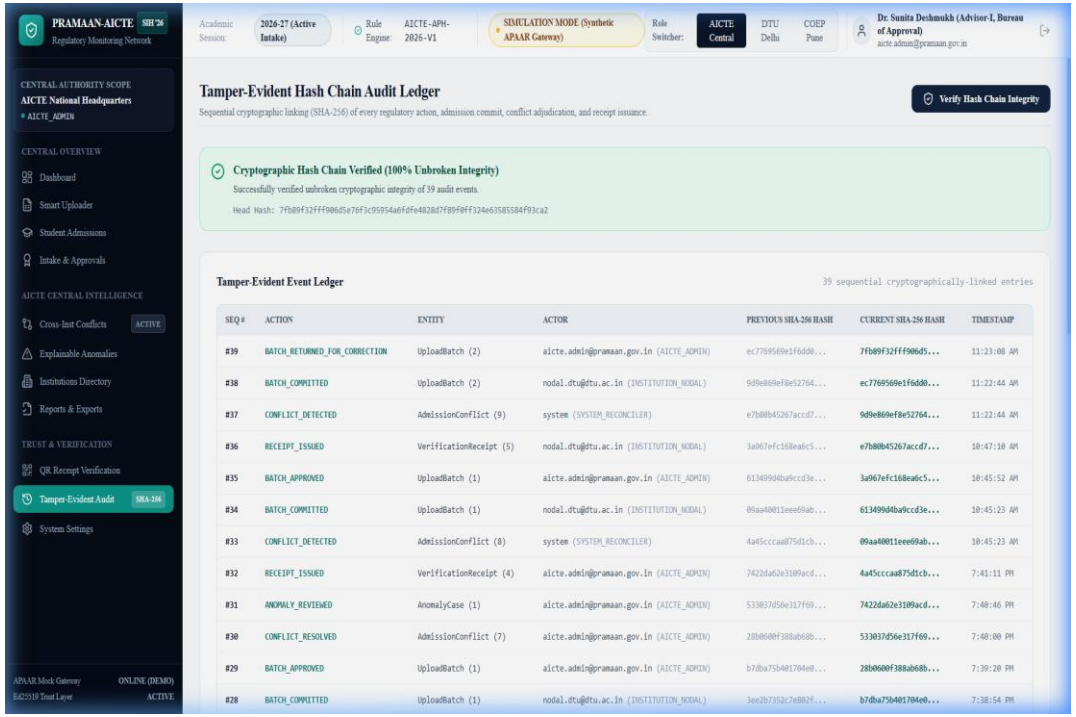
RESEARCH CITATIONS, REPOSITORY & LIVE EVALUATION LINKS

Formal Regulatory & Cryptographic References

- **AICTE APH 2026-27:** Approval Process Handbook norms for sanctioned intake ceilings, supernumerary quotas, and NRI/OCI/FN rules.
- **NEP 2020 & NAD:** Framework for APAAR (Automated Permanent Academic Account Registry) and Academic Bank of Credits (ABC).
- **IETF RFC 8032:** Edwards-Curve Digital Signature Algorithm (Ed25519 / Curve25519) for tamper-proof academic credentials.
- **DPDP Act 2023:** Section 4 data fiduciary mandates, purpose limitation, identifier pseudonymization, and student consent mechanisms.
- **NIST SP 800-107 & Verhoeff:** Permutation algorithms on Dihedral Group D5 for checksums and sequential SHA-256 cryptographic hash-chains.

Live Deployment, Source Code & Verification Links

- **Open-Source GitHub Repo:** https://github.com/naveeshpcbin/APAAR-ID-BASED-SYSTEM
- **Live Hosted Platform:** https://pramaan-aicte.yourdomain.com (Paste custom domain here)
- **Interactive Swagger Docs:** https://pramaan-backend.onrender.com/docs (FastAPI OpenAPI 3.1)
- **Supabase Cloud Database:** Ref: xvuqedqfdeybryvbablt (PostgreSQL Pooler aws-0-ap-south-1)
- **Evaluator Quick Credentials:** Central Admin: aicte.admin@pramaan.gov.in | Password: Admin@12345



PROTOTYPE EVIDENCE & LIVE VERIFICATION

Tamper-Evident SHA-256 Cryptographic Audit Ledger (/audit) verifying unbroken hash-chain continuity from Genesis Block 0 across all admissions, batch commits, and review decisions.